Bayesian augmentation by chained equations for phylogenetic comparative data: the BACE R package

Abstract

1. Missing trait data are pervasive in comparative datasets limiting the types of questions that can be tested and compromising the power of comparative analyses. Standard approaches, such as complete-case analyses and single imputation, can bias inference and underestimate uncertainty, particularly when missing data is non-random. While robust imputation methods exits, many existing software are restricted to imputing only continuous or count variables and lack the capacity to impute discrete / categorical traits (e.g., multinomial, ordered categorical), which are common trait types in comparative datasets. Such limitations have prevented the widespread use of imputation approaches in comparative biology.
2. We introduce *BACE*, an R package implementing Bayesian Augmentation by Chained Equations for phylogenetic comparative data. *BACE* enables imputation with phylogeny-informed covariance that can account for both phylogenetic and non-phylogenetic species effects, along with mixed data types, in a fully Bayesian pipeline. *BACE* provides substantial flexibility in the imputation models fit, can decouple imputation models from analysis models, and propagates both parameter and imputation uncertainty in subsequent downstream analyses.
3. We show using simulations that *BACE* recovers missing trait values with a high degree of accuracy. Across 18 different scenarios that vary in sample size (n = 100, 300, 1000 species), phylogenetic signal (Pagel's = 0.3, 0.7, 1) and trait correlation (= 0, 0.5), we show *BACE* recovers model parameters with little bias and nominal coverage. *BACE* outperformed other comparable frequentist approaches.
4.

Introduction

Phylogenetic comparative methods (PCMs) provide a framework for understanding how traits evolve and covary across species while accounting for their shared evolutionary history (Felsenstein, 1985; Grafen, 1989; Martins & Hansen, 1997; Pagel, 1994). Expanding trait databases, increasingly comprehensive phylogenies, and greater access to environmental data provide opportunities to investigate how evolutionary history and environmental conditions shape variation and covariation among traits allowing one to test a rich set of evolutionary hypotheses (Blomberg & Todorov, 2025; Halliwell et al., 2025). However, missing trait data limit the questions that can be addressed, and excluding species with incomplete records can reduce sample sizes, statistical power and precision (Blomberg & Todorov, 2025; Little et al., 2024; Nakagawa & Freckleton, 2008). When the availability of trait data depends on species' characteristics, this exclusion can also bias estimates of trait relationships (González-Suárez et al., 2012; Nakagawa & Freckleton, 2008). Empirical studies illustrate the consequences of missing data for answering questions in comparative biology. For example, incomplete trait coverage altered inferred relationships between mammalian traits and extinction risk (González-Suárez et al., 2012), while alternative treatments of missing observations have been shown to change estimates of reproductive performance and temporal trends in ecological monitoring (Łopucki et al., 2022).

Methods for handling missing data have a long history in statistics (Schafer & Graham, 2002). Approaches developed for comparative datasets include phylogenetic prediction of missing trait values in PhyloPars and Rphylopars (Bruggeman et al., 2009; Goolsby et al., 2017), as well as multiple-imputation procedures that account for phylogenetic and species-sampling uncertainty (Blomberg & Todorov, 2025; Nakagawa & de Villemereuil, 2018). Missing values can be accommodated within model fitting, for example through Bayesian data augmentation, which samples missing values alongside model parameters (Tanner & Wong, 1987). Alternatively, missing data can be handled in a multiple-imputation workflow, where m complete datasets are generated and analysed separately. Parameter estimates and their variances are then combined, commonly using Rubin's rules, to account for both within- and between-imputation uncertainty (Noble & Nakagawa, 2021; Rubin, 1987). Data augmentation can also be used to generate the completed datasets for multiple imputation (Schafer & Graham, 2002). In chained-equations approaches, conditional imputation models must also be specified carefully, because incompatible specifications can bias parameter estimates (Du et al., 2022).

Recent work has also addressed the imputation of categorical traits. Gendre et al. (2024) benchmarked phylogenetic, machine-learning and ensemble approaches, combining phylogenetic predictions with information from other traits in the TDIP package. Their evaluation focused on the accuracy of imputed categorical states. Accurate prediction alone, however, does not establish valid downstream inference: treating a single completed dataset as fully observed can understate uncertainty in model parameters (Blomberg & Todorov, 2025). This distinction motivates evaluating parameter bias and interval coverage in downstream analyses alongside the accuracy of imputed values.

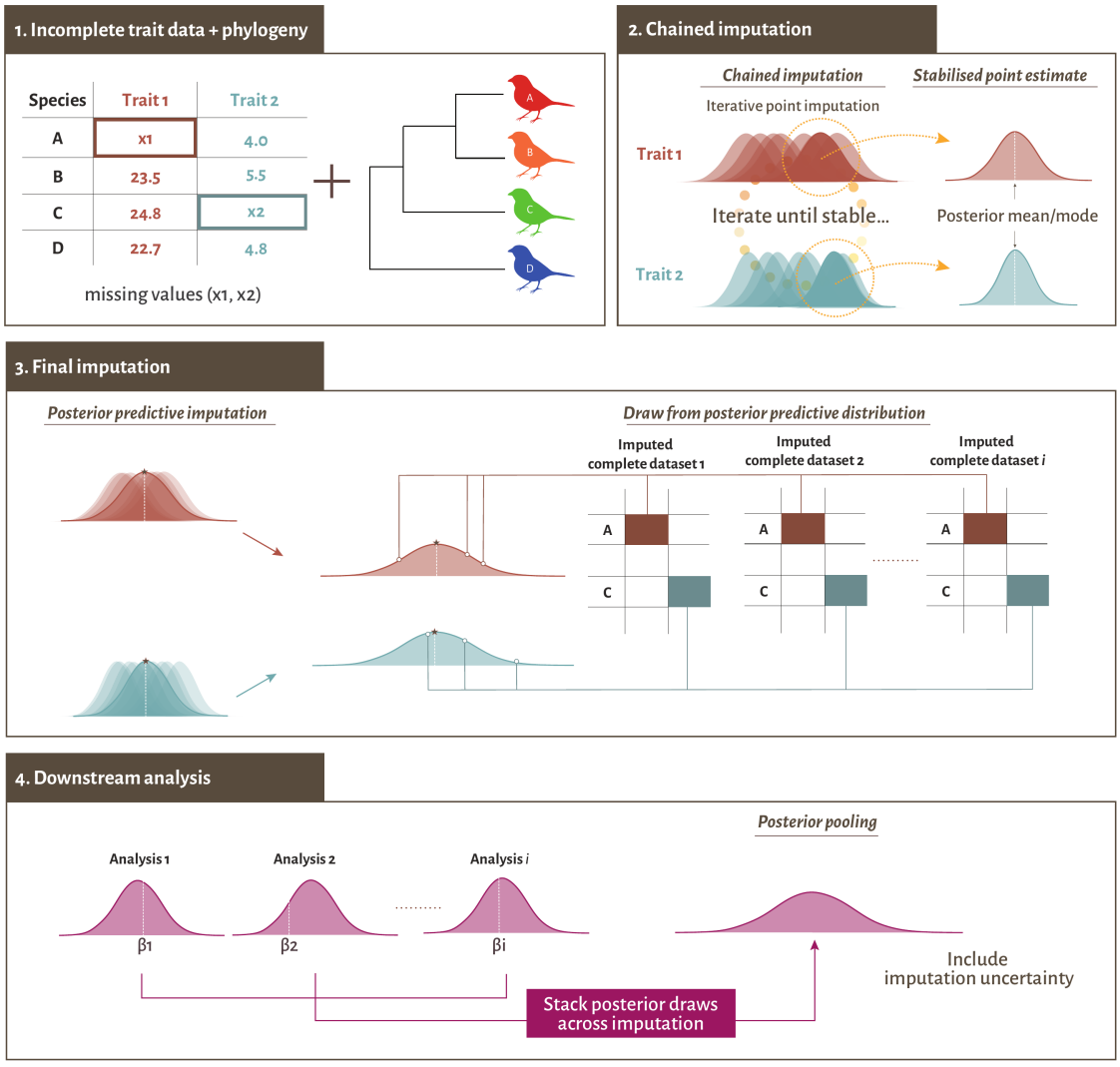


Figure 1: **Overview of the BACE imputation pipeline**. (1) The inputs are a comparative dataset with missing trait values and a phylogeny covering the species. (2) Chained imputation: each variable with missing data gets its own phylogenetic mixed model, and the models are cycled iteratively — each iteration replacing the missing values with point estimates (posterior mean/mode) — until the imputed values stabilise. (3) Final imputation: starting from the converged dataset, each missing value is drawn from its posterior predictive distribution, independently and repeatedly, yielding multiple completed datasets that together capture imputation uncertainty. (4) Posterior distributions pooled: If the analysis model is included in the imputation set then inferences can be done on the pooled posterior which pools uncertainty within a model fit to the compete data as well as betwene imputation uncertainty; (5) Further downstream analysis: alternatively, a different analysis model can be fit to every completed dataset and the posterior draws stacked (pooled) across imputations (if Bayesian) or parameter estimates and uncertainty pooled using Rubin's rules. In this way, the final posterior reflects both parameter and imputation uncertainty.

Missing data in comparative biology: why it matters

- Trait databases + environmental covariates + phylogenies → systematic missingness (taxonomic, geographic, trait-type).
- Consequences: reduced power, compositional bias (which taxa remain), and overconfident inference if uncertainty is ignored.
- comparative data in conservation biology, macroecology, evolutionary biology, functional ecology. Used to inform theory, policy, and management.

Why multiple imputation, and why *Bayesian* chained equations

- MI as uncertainty propagation (imputation as inference, not a preprocessing "fill").
- Chained equations: flexible conditional models per variable; practical for mixed data types and modular model building.
- Bayesian approach: coherent uncertainty, hierarchical structure, prior regularisation, posterior predictive checks.

Gap in current tooling (positioning)

Table 1 summarises the R tools available for imputing missing trait data, against the three requirements of comparative analyses: respecting shared evolutionary history, handling the mixed variable types that trait datasets contain, and propagating imputation uncertainty into downstream inference. Every capability shown was verified against the package's own reference documentation (current CRAN or GitHub version, October 2026).

Two gaps stand out. First, no existing tool combines a phylogeny with proper multiple imputation. The general multiple-imputation frameworks — *mice* and its multilevel extensions, the joint-modelling packages, *JointAI* and *smcfcs* — propagate imputation uncertainty correctly but represent grouping only as exchangeable cluster effects: none can encode a covariance structure derived from a tree (*JointAI*'s random-effect covariance options, for example, are limited to full, block-diagonal or independent; Erler et al. (2021)). The phylogeny-aware tools, conversely, either return a single completed dataset or point predictions (*Rphylopars*, *TDIP*, *BHPMF*) or impute only continuous traits (*Rphylopars*, *phylomice*, *BHPMF*), and several are unmaintained or unpublished. Second, coverage of variable types is thinner than reputations suggest: *mice* has no count method — its documented defaults cover numeric, binary, unordered-factor and ordered-factor variables only (Buuren & Groothuis-Oudshoorn, 2011) — and the widely used multilevel route in *miceadds* cannot impute nominal variables at all. The only packages covering all five variable types in Table 1 are *smcfcs* (single-level only), *JointAI* (no phylogeny), and *hmi*, which was removed from CRAN in 2022.

BACE fills the empty cell: chained-equations multiple imputation in which every conditional model is a phylogenetic mixed model, supporting continuous, count, binary, ordinal and nominal traits, accommodating replicated species, and propagating both parameter and imputation uncertainty into downstream analyses.

Table 1: R tools for imputing missing data, against the needs of phylogenetic comparative analysis. *Phylogeny*: a tree-derived covariance informs the imputation. *Proper MI*: multiple imputed datasets (or an integrated Bayesian posterior) are produced, so downstream analyses reflect imputation uncertainty. *Multilevel*: non-independence beyond the phylogeny (random effects / clustered rows). Variable-type columns: an explicit imputation model for continuous, count, binary, ordinal (treated as ordered) and nominal (unordered multinomial) variables. ✓ = documented in the package's reference documentation (October 2026); (✓) = partial (see notes); ✗ = not documented. Column abbreviations: Phylo. = phylogeny; MI = proper multiple imputation; Multi. = multilevel; Cont. = continuous; Bin. = binary; Ord. = ordinal; Nom. = nominal.

Package	Phylo.	MI	Multi.	Cont.	Count	Bin.	Ord.	Nom.	Reference
BACE	✓	✓	(✓)[a]	✓	✓	✓	✓	✓	this paper
Rphylopars	✓	✗[b]	(✓)[c]	✓	✗	✗	✗	✗	Goolsby et al. (2017)
phylomice	✓	✓	✗	✓	✗	✗	✗	✗	Drhlik (2016)[d]
TDIP	✓	✗[e]	✗	✓	✗	✓	✗[f]	✓	Gendre et al. (2024)[d]
BHPMF	(✓)[g]	✗[h]	(✓)[g]	✓	✗	✗	✗	✗	Schrodt et al. (2015)[i]
mice	✗	✓	(✓)[j]	✓	✗[k]	✓	✓	✓	Buuren & Groothuis-Oudshoorn (2011)
miceadds	✗	✓	✓	✓	✗	✓	(✓)[l]	✗[l]	Robitzsch & Grund (2026)
micemd	✗	✓	✓	✓	✓	✓	(✓)[m]	(✓)[m]	Audigier et al. (2018)
pan	✗	✓	✓	✓	✗	✗	✗	✗	Schafer & Yucel (2002)
jomo	✗	✓	✓	✓	✗[n]	✓	(✓)[n]	✓	Carpenter & Kenward (2013)
mitml	✗	✓	✓	✓	✗	✓	✗[o]	✓	Grund et al. (2023)[o]
hmi	✗	✓	✓	✓	✓	✓	✓	✓	Speidel et al. (2020)[p]

Package	Phylo.	MI	Multi.	Cont.	Count	Bin.	Ord.	Nom.	Reference
mdmb	✗	✓	(✓)[q]	✓	✗	✓	✓	✗[r]	Robitzsch & Lüdtke (2024)
JointAI	✗	(✓)[s]	✓	✓	✓	✓	✓	✓	Erler et al. (2021)
smcfcs	✗	✓	✗	✓	✓	✓	✓	✓	Bartlett et al. (2015)
missForest	✗	✗[t]	✗	✓	✗	(✓)[u]	✗	✓	Stekhoven & Bühlmann (2012)

[a] Phylogenetic random effect plus an optional non-phylogenetic species effect for species with replicated rows; arbitrary further grouping factors are not supported. [b] Single predicted value with a prediction variance per missing cell. [c] Multiple observations per species (within-species variation), not general grouping factors. [d] Not on CRAN (GitHub only); *phylomice* is experimental and last updated in 2016; *TDIP*'s documentation cites no publication (we cite the authors' benchmark paper). [e] Returns single completed datasets (including a majority-vote ensemble); no pooling or between-imputation variance. [f] "Ordinal" appears only as a simulated trait class; the imputation methods document unordered polytomous regression and generic discrete models. [g] Hierarchy is taxonomic (species/genus/family), not a phylogeny with branch lengths. [h] Single imputed matrix with a per-cell standard deviation. [i] Archived from CRAN in 2017. [j] Two-level normal and logistic methods only. [k] No count method: the documented defaults cover numeric, binary, unordered-factor and ordered-factor variables only. [l] The multilevel routine states "Nominal variables with missing values cannot (yet) be handled"; ordinal variables are imputed by predictive mean matching rather than an ordinal model. [m] No two-level ordinal or nominal methods; falls back to *mice*'s single-level proportional-odds and polytomous methods. [n] Imputation families are gaussian/binomial only; ordered models appear only as substantive-model-compatible outcomes (`jomo.polr`, `jomo.clmm`) — the general joint model treats categorical variables as unordered. [o] Front-end to *pan* and *jomo*; type support is inherited, and ordered factors are not distinguished. [p] Removed from CRAN in 2022; its pooling helper averages estimates without variances and defers to `mice::pool`. [q] Multilevel models in the fully Bayesian estimation path only. [r] Nominal covariate models "will be included in future mdmb package versions". [s] Imputation is integrated with the analysis model (fully Bayesian); multiply imputed datasets can be exported (`get_MIdat()`). [t] Single completed dataset; the out-of-bag error estimates accuracy, not imputation uncertainty. [u] Factors via classification forests; ordered factors are not handled as ordered.

Overview of the BACE framework

BACE provides options for direct imputation and parameter recovery of a given model with missing trait data or imputed trait data which can be pooled in downstream analyses such that the imputation and analysis models are different. ## Concept: Bayesian augmentation by chained equations - Describe the iterative cycle (variable-by-variable conditional models). - Emphasise "augmentation": draw missing values from the posterior predictive distribution each cycle.

Phylogenetic information in the imputation model

- How the tree enters (e.g., covariance among species; phylogenetic random effect).
- How this interacts with species-level random effects and other grouping structures.

Supported data types and models (v1 scope)

(Explicitly list what is supported in BACE v1 to avoid overclaiming.) - Continuous - Binary - Count - Ordered categorical - Unordered categorical (multinomial) *(if/when supported)* - Missingness patterns handled (MCAR/MAR; sensitivity options if present)

> **Design choice box (optional):** When to use a joint-model vs chained equations; why BACE chooses chained equations.

The BACE R package

Inputs, outputs, and core workflow

- Inputs: data frame, species ID, phylogeny, optional predictors, grouping factors, priors, M (number of imputations / posterior draws).
- Outputs: imputed datasets (long/wide), posterior summaries, diagnostics, helper functions for pooling and downstream analysis.

Typical user pipeline (pseudo-code)

```r
# 1) Prepare data and tree (rows = taxa; factors declared; counts as integers)
# 2) Run the full workflow: chained imputation -> convergence check ->
#    n_final posterior-predictive imputations -> pooled posteriors
res <- bace(fixformula     = "y ~ x1 + x2",
            ran_phylo_form = "~ 1 | Species",
            phylo          = tree,
            data           = dat,
            runs           = 10, n_final = 50)

# 3) Diagnostics
plot(assess_convergence(res$initial_results))

# 4a) Bayesian route: pooled MCMCglmm posteriors
summary(get_pooled_model(res, "y"))

# 4b) Rubin's-rules route: any downstream model per imputed dataset
fits <- with_imputations(res, function(d) lm(y ~ x1 + x2, data = d))
pool_mi(fits)
```

Convergence diagnostics

(Structure only — content to write. See tutorial "Two kinds of convergence".) - Within-chain MCMC convergence of each per-variable model vs convergence of the chained-equations loop (the imputed-data sequence). - `assess_convergence()` criteria (%-change + trend + ACF) and the `plot_*_convergence()` family.

Pooling pathways: stacked posteriors and Rubin's rules

(Structure only. Sources: `.agents/plans/pool_mi_rubin.md`; tutorial "Interpreting the pooled output"; Zhou & Reiter 2010; BDA3 Ch. 18.) - `pool_posteriors()`: stacking per-imputation chains approximates the marginal posterior; not Rubin's rules; O(1/M) difference at small M. - `with_imputations()` + `pool_mi()`: model-agnostic Rubin's rules (Barnard–Rubin df, fmi/riv); matches `mice::pool` to machine precision. - When to use which (variance components never via Rubin).

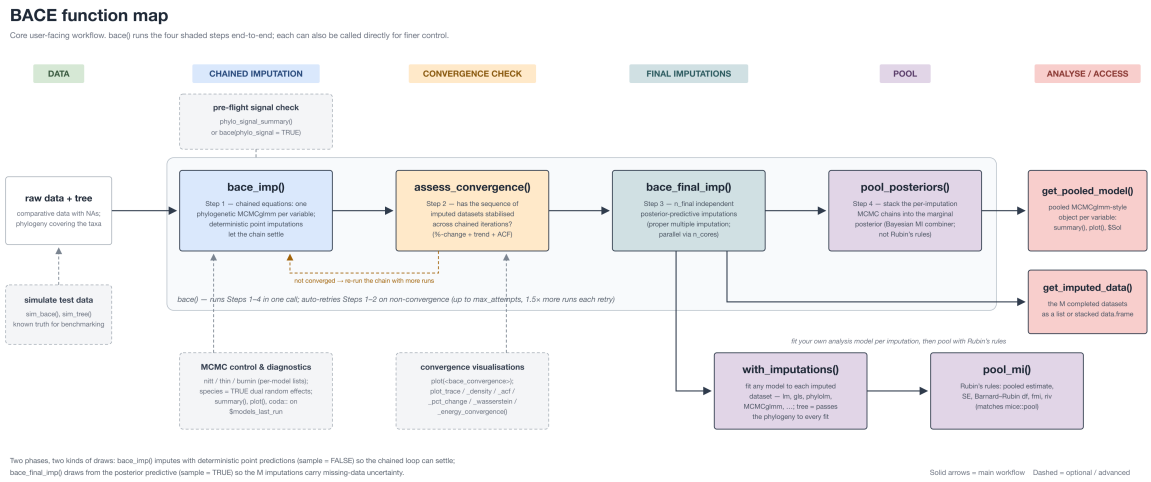


Figure 2: *(Caption to write.)* Core user-facing BACE function map.

Performance evaluation

Simulation design

(Structure only. Sources: `dev/12_recovery_simulation.R`*,* `dev/13_response_type_simulation.R`*,* `dev/simulation_results/SIMULATION_REPORT.qmd`*; design details in* `.agents/simulation-report.html`*.)*

- Study A: parameter recovery (gaussian slope) under MCAR/MAR; oracle and complete-case comparators. - Study B: five response types x MCAR/MAR; per-type accuracy/coverage metrics and their finite-draw/metric ceilings (`cover95_t`, prob-based sets, Bayes ceiling for categorical).

Parameter recovery and inferential calibration (Study A)

Imputation accuracy across response types (Study B)

Benchmark against existing tools

(Pending Track B3: missForest, Rphylopars, mice as feasible. Note: pigauto is a companion package, not a competitor — see `.agents/pigauto.md` *for positioning.)*

Worked example

(Pending Track B4: AVONET at production MCMC settings; 300-species subset exists from the April benchmark.)

Discussion

(Structure only.) - What BACE adds relative to existing MI and phylogenetic-prediction tools. - Honest limitations - Practical guidance (pre-flight signal screening; when BACE is the wrong tool). - Relationship to pigauto (division of labour).

Availability and reproducibility

(Structure only.) - Package: GitHub (daniel1noble/BACE), tutorial site, CRAN (pending). - Simulation reproduction: `dev/12`, `dev/13`, `dev/14`, `SIMULATION_REPORT.qmd` (commands in `.agents/README.md`).

Acknowledgements

(To write.)

Author contributions

(To write — CRediT.)

Data and code availability

(To write.)

References

Audigier, V., White, I. R., Jolani, S., Debray, T. P. A., Quartagno, M., Carpenter, J., Buuren, S. van, & Resche-Rigon, M. (2018). Multiple imputation for multilevel data with continuous and binary variables. *Statistical Science*, *33*(2), 160–183. https://doi.org/10.1214/18-STS646

Bartlett, J. W., Seaman, S. R., White, I. R., & Carpenter, J. R. (2015). Multiple imputation of covariates by fully conditional specification: Accommodating the substantive model. *Statistical Methods in Medical Research*, *24*(4), 462–487. https://doi.org/10.1177/0962280214521348

Blomberg, S. P., & Todorov, O. S. (2025). The fallacy of single imputation for trait databases: Use multiple imputation instead. *Methods in Ecology and Evolution*, *16*(4), 658–667. https://doi.org/10.1111/2041-210X.14494

Bruggeman, J., Heringa, J., & Brandt, B. W. (2009). PhyloPars: Estimation of missing parameter values using phylogeny. *Nucleic Acids Research*, *37*(suppl_2), W179–W184. https://doi.org/10.1093/nar/gkp370

Buuren, S. van, & Groothuis-Oudshoorn, K. (2011). Mice: Multivariate imputation by chained equations in r. *Journal of Statistical Software*, *45*(3), 1–67. https://doi.org/10.18637/jss.v045.i03

Carpenter, J. R., & Kenward, M. G. (2013). *Multiple imputation and its application*. Wiley. https://doi.org/10.1002/9781119942283

Drhlik, P. (2016). *Phylomice: Phylogenetic imputation methods for mice*. https://github.com/pdrhlik/phylomice

Du, H., Alacam, E., Mena, S., & Keller, B. T. (2022). Compatibility in imputation specification. *Behavior Research Methods*, *54*(6), 2962–2980. https://doi.org/10.3758/s13428-021-01749-5

Erler, N. S., Rizopoulos, D., & Lesaffre, E. M. E. H. (2021). JointAI: Joint analysis and imputation of incomplete data in r. *Journal of Statistical Software*, *100*(20), 1–56. https://doi.org/10.18637/jss.v100.i20

Felsenstein, J. (1985). Phylogenies and the comparative method. *The American Naturalist*, *125*(1), 1–15. https://doi.org/10.1086/284325

Gendre, M., Hauffe, T., Pimiento, C., & Silvestro, D. (2024). Benchmarking imputation methods for categorical biological data. *Methods in Ecology and Evolution*, *15*(9), 1624–1638. https://doi.org/10.1111/2041-210X.14339

González-Suárez, M., Lucas, P. M., & Revilla, E. (2012). Biases in comparative analyses of extinction risk: Mind the gap. *Journal of Animal Ecology*, *81*(6), 1211–1222. https://doi.org/10.1111/j.1365-2656.2012.01999.x

Goolsby, E. W., Bruggeman, J., & Ané, C. (2017). Rphylopars: Fast multivariate phylogenetic comparative methods for missing data and within-species variation. *Methods in Ecology and Evolution*, *8*(1), 22–27. https://doi.org/10.1111/2041-210X.12612

Grafen, A. (1989). The phylogenetic regression. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, *326*(1233), 119–157. https://doi.org/10

.1098/rstb.1989.0106
Grund, S., Robitzsch, A., & Lüdtke, O. (2023). *Mitml: Tools for multiple imputation in multilevel modeling*. https://CRAN.R-project.org/package=mitml
Halliwell, B., Holland, B. R., & Yates, L. A. (2025). Multi-response phylogenetic mixed models: Concepts and application. *Biological Reviews of the Cambridge Philosophical Society*, *100*(3), 1294–1316. https://doi.org/10.1111/brv.70001
Little, R. J., Carpenter, J. R., & Lee, K. J. (2024). A comparison of three popular methods for handling missing data: Complete-case analysis, inverse probability weighting, and multiple imputation. *Sociological Methods & Research*, *53*(3), 1105–1135. https://doi.org/10.1177/00491241221113873
Łopucki, R., Kiersztyn, A., Pitucha, G., & Kitowski, I. (2022). Handling missing data in ecological studies: Ignoring gaps in the dataset can distort the inference. *Ecological Modelling*, *468*, 109964. https://doi.org/10.1016/j.ecolmodel.2022.109964
Martins, E. P., & Hansen, T. F. (1997). Phylogenies and the comparative method: A general approach to incorporating phylogenetic information into the analysis of interspecific data. *The American Naturalist*, *149*(4), 646–667. https://doi.org/10.1086/286013
Nakagawa, S., & de Villemereuil, P. (2018). A general method for simultaneously accounting for phylogenetic and species sampling uncertainty via rubin's rules in comparative analysis. *Systematic Biology*, *68*(4), 632–641. https://doi.org/10.1093/sysbio/syy089
Nakagawa, S., & Freckleton, R. P. (2008). Missing inaction: The dangers of ignoring missing data. *Trends in Ecology & Evolution*, *23*(11), 592–596. https://doi.org/10.1016/j.tree.2008.06.014
Noble, D. W. A., & Nakagawa, S. (2021). Planned missing data designs and methods: Options for strengthening inference, increasing research efficiency and improving animal welfare in ecological and evolutionary research. *Evolutionary Applications*, *14*(8), 1958–1968. https://doi.org/10.1111/eva.13273
Pagel, M. (1994). Detecting correlated evolution on phylogenies: A general method for the comparative analysis of discrete characters. *Proceedings of the Royal Society of London. Series B: Biological Sciences*, *255*(1342), 37–45. https://doi.org/10.1098/rspb.1994.0006
Robitzsch, A., & Grund, S. (2026). *Miceadds: Some additional multiple imputation functions, especially for mice*. https://CRAN.R-project.org/package=miceadds
Robitzsch, A., & Lüdtke, O. (2024). *Mdmb: Model based treatment of missing data*. https://CRAN.R-project.org/package=mdmb
Rubin, D. B. (1987). *Multiple imputation for nonresponse in surveys*. John Wiley & Sons. https://doi.org/10.1002/9780470316696
Schafer, J. L., & Graham, J. W. (2002). Missing data: Our view of the state of the art. *Psychological Methods*, *7*(2), 147–177. https://doi.org/10.1037/1082-989X.7.2.147
Schafer, J. L., & Yucel, R. M. (2002). Computational strategies for multivariate linear mixed-effects models with missing values. *Journal of Computational and Graphical Statistics*, *11*(2), 437–457. https://doi.org/10.1198/106186002760180608
Schrodt, F., Kattge, J., Shan, H., Fazayeli, F., Joswig, J., Banerjee, A., Reichstein, M., Bönisch, G., Díaz, S., Dickie, J., Gillison, A., Karpatne, A., Lavorel, S., Leadley, P., Wirth, C. B., Wright, I. J., Wright, S. J., & Reich, P. B. (2015). BHPMF – a hierarchical bayesian

approach to gap-filling and trait prediction for macroecology and functional biogeography. *Global Ecology and Biogeography*, *24*(12), 1510–1521. https://doi.org/10.1111/geb.12335

Speidel, M., Drechsler, J., & Jolani, S. (2020). The r package hmi: A convenient tool for hierarchical multiple imputation and beyond. *Journal of Statistical Software*, *95*(9), 1–48. https://doi.org/10.18637/jss.v095.i09

Stekhoven, D. J., & Bühlmann, P. (2012). MissForest—non-parametric missing value imputation for mixed-type data. *Bioinformatics*, *28*(1), 112–118. https://doi.org/10.1093/bioinformatics/btr597

Tanner, M. A., & Wong, W. H. (1987). The calculation of posterior distributions by data augmentation. *Journal of the American Statistical Association*, *82*(398), 528–540. https://doi.org/10.1080/01621459.1987.10478458